Image Enhancement

Computer Vision - Homework 1

Alberto Spada VR541032

1 Introduction

This report presents a methodology to enhance and restore degraded images using classical computer vision techniques available in Python via libraries such as *OpenCV* and *NumPy*. The main objective of this work is to compare, both quantitatively and qualitatively, the results of the applied methods. Specifically, this analysis aims to observe the effects of different spatial domain operations and evaluate the trade-offs among detail visibility, noise amplification, and natural appearance.
The repository for this project is available at the following address: `https://github.com/Lambepro1/Computer-Vision---Homework-1`

2 Implementation

2.1 Dataset

The dataset consists of 5 color images exhibiting varying degrees of illumination challenges.

Image	**Visual effects**
Img 0	low contrast, bad illumination, poorly visible details
Img 1	non-uniform illumination, shadows
Img 2	poorly visible details, non-uniform illumination
Img 3	low contrast
Img 4	low contrast

Table 1: Characteristics of the sample images

To generate the dataset for the restoration pipeline, the original images were synthetically degraded by adding a small amount of Gaussian blur and decreasing their global contrast. The original, uncorrupted images serve strictly as ground-truth references for evaluating the synthetic degradations, as a clean reference may not be available in real-world scenarios. All original images were taken by the author and are used under a custom license.

2.2 Methodology

Three different approaches were deployed to restore the degraded images:

- **Linear Contrast Stretching** is a point operation that expands the narrow range of intensity levels in the degraded image to span the full dynamic range $[0, 255]$. In our implementation, min-max normalization is applied. This operation was performed independently on each color channel (Red, Green, and Blue).
- Standard global histogram equalization often leads to over-amplification of noise, which is heavily noticeable when applying standard normalization on HSV channels. To mitigate

this, **Contrast Limited Adaptive Histogram Equalization (CLAHE)** was utilized. To preserve the original colors, the image was first converted from the *BGR* to the *HSV* color space. CLAHE was then applied exclusively to the Value (V) channel (luminance). Finally, the image was converted back to the *BGR* color space.

- As a fourth method, we implemented an edge-preserving smoothing approach using the **Bilateral Filter + Saturation enhancement**. The bilateral filter replaces the intensity of each pixel with a weighted average of intensity values from nearby pixels, ensuring that strong edges are maintained while flat, noisy regions are smoothed. After applying the filter, the image was converted to *HSV*. The Saturation (S) channel was mildly boosted to restore color vividness lost during the smoothing phase, and the result was merged back.

2.3 Evaluation

Below are the numerical and qualitative results. Two metrics were used to evaluate the restoration performance against the original ground-truth images: **PSNR** (Peak Signal-to-Noise Ratio) states how close the restored pixels are to the original image (a good value should generally be above 30 dB). **SSIM** (Structural Similarity Index) yields a value between 0 and 1, measuring how similar the image structure (edges, textures, luminance) looks to the human eye.

Image	**PSNR**	**SSIM**
Img 0	23.8365	0.5539
Img 1	28.2161	0.8451
Img 2	34.2865	0.9552
Img 3	15.9542	0.6697
Img 4	26.5663	0.7261

Table 2: Results of Linear Contrast Stretching

Image	**PSNR**	**SSIM**
Img 0	20.2577	0.5013
Img 1	19.7868	0.5347
Img 2	20.8512	0.7447
Img 3	21.6194	0.7360
Img 4	15.0876	0.6850

Table 3: Results of CLAHE

Image	**PSNR**	**SSIM**
Img 0	20.8092	0.4275
Img 1	20.6038	0.5474
Img 2	18.5298	0.7320
Img 3	26.5741	0.7289
Img 4	13.8820	0.6085

Table 4: Results of Bilateral Filter + Saturation

Figure 1: Results of img0. From left to right: Original, Degraded, Linear Stretching, CLAHE, Bilateral Filter.

Figure 2: Results of img1. From left to right: Original, Degraded, Linear Stretching, CLAHE, Bilateral Filter.

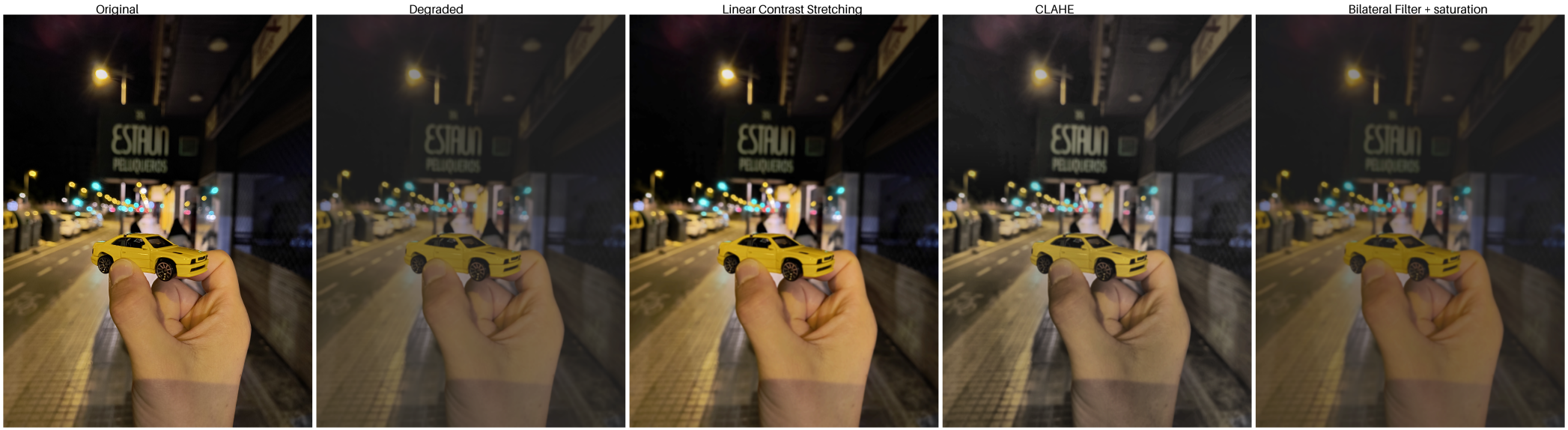

Figure 3: Results of img2. From left to right: Original, Degraded, Linear Stretching, CLAHE, Bilateral Filter.

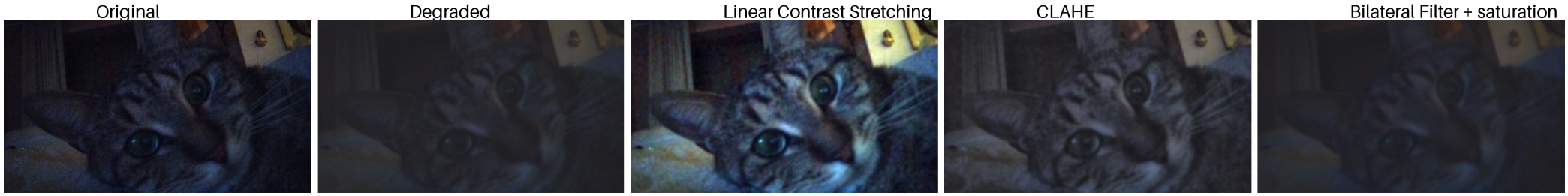

Figure 4: Results of img3. From left to right: Original, Degraded, Linear Stretching, CLAHE, Bilateral Filter.

Figure 5: Results of img4. From left to right: Original, Degraded, Linear Stretching, CLAHE, Bilateral Filter.

3 Considerations and Conclusions

Looking at the images and the scores, it is possible to make some general considerations that are valid for the whole dataset.

Linear Contrast Stretching actually smashed the metrics, obtaining the highest PSNR and SSIM scores. This makes sense because it mathematically reverses the contrast reduction we used to degrade the images. Even qualitatively, the images seem to be similar to the original ones.

CLAHE is the method that stands in the middle. It obtained lower numerical scores, but visually it gives the better results. It brings out way more details in the shadows, and it is easier for the human eye to understand the scene. The metrics penalize it just because it alters the original lighting, showing a clear case where a worse numerical score still gives a better visual feeling.

To conclude, image restoration with the third method (Bilateral Filter + Saturation) will surely obtain better results in edge detection tasks, because the smooth edges and flat areas are clearer to identify by a classifier. But, the images produced are the worst in terms of fine details, resolution, and capability to express textures, which is why it failed when the metrics were computed. In short, Linear Stretching gives us the best mathematical goodness for the restoration task, but CLAHE is the clear winner for human perception.